

Revised Generation Interconnection System Impact Study Report

For Queue Project Q21-1760

Flambeau Hydroelectric Station 230 kV
August 22, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q21-1760. The Interconnection Customer proposes a 132.5 MW battery energy storage facility interconnecting at the Flambeau Hydroelectric Station 230 kV substation in the Illinois area. The request was received on May 26, 2021. The study evaluated system intact and single contingency conditions using the regional planning case, including 560 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 3 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,555,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Ft St Vrain / University Of Colorado 230.0 kV L
Contingency: Flambeau Hydroelectric Sta / Pueblo Xtra 230.0 kV
Loading: 129.4% (Rating: 567 MVA)
DFAX: 53.0%
Pre-project loading: 117.0%
2. Facility: Flambeau Hydroelectric Sta / Pueblo Xtra 230.0 kV
Contingency: University Of Colorado / Ft St Vrain 230.0 kV
Loading: 116.6% (Rating: 737 MVA)
DFAX: 82.2%
Pre-project loading: 101.8%
3. Facility: Liggett / E Frankfort Substation 100.0 kV line
Contingency: Flambeau Hydroelectric Station Substation / Pueblo Xtra 230.0 kV Line
Loading: 114.9% (Rating: 113 MVA)
DFAX: 5.9%
Pre-project loading: 107.9%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
University Of Colorado / Ft St Vrain 230.0 kV	Rebuild line with higher-capacity conductor	\$32,826,000	14% (\$4,739,000)
Flambeau Hydroelectric Sta / Pueblo Xtra 230.0 kV	Rebuild line with higher-capacity conductor	\$25,414,000	14% (\$3,645,000)
Liggett / E Frankfort Substation 100.0 kV line	Rebuild line with higher-capacity conductor	\$3,142,000	5% (\$171,000)
	Total allocated to project		\$8,555,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.